AEROSPACE INFORMATION RESEARCH INSTITUTE
Chinese Academy of Sciences
State Key Laboratory of Remote Sensing and Digital Earth

Drought Event Analysis in South Punjab, Pakistan

Multi-Sensor Remote Sensing of Precipitation, Soil Moisture, Vegetation and Thermal Cascades, and Exploratory Implications for Wheat Yield

Analysis Report for Research Application

Topic: Drought Event Analysis | Satellite Remote Sensing for Water Resources & Climate Change

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Abstract

Eleven districts of South Punjab, Pakistan were assembled into a monthly, district-level multi-sensor panel covering 2001-2025: CHIRPS precipitation, ERA5-Land soil water, MODIS Terra and Aqua land-surface temperature, MODIS NDVI, and MODIS ET. Standardized precipitation and soil-water anomalies (SPI-1/SPI-3 and a soil-moisture z-score) identify drought events and their duration and severity, and lag analysis tests how precipitation deficit propagates into soil-water, thermal, vegetation, and evapotranspiration responses. A static canal-exposure proxy — the share of each district's cropland within 5 km of mapped OSM canals — captures spatial variation in irrigation-infrastructure proximity; its interaction with the drought metrics is tested against a district-year panel of official wheat-yield residuals (harvest years 2001-2025).

At the primary threshold of -1.0 standard deviations, the eleven districts recorded 91 SPI-1 events and 48 soil-moisture events. All four tested lag relationships — precipitation to soil water, and soil water to thermal, vegetation and ET anomalies — peak at a contemporaneous, zero-month lag and hold up as significant in every district after FDR correction; the association weakens in subsequent months rather than building toward a later peak. Canal exposure itself varies widely across the study area, from 0.45 in Layyah to 0.93 in Bahawalnagar. None of the three pre-specified fixed-effects regressions of wheat-yield residuals on Rabi-season drought metrics (soil moisture, SPI-1, SPI-3) and their interaction with canal exposure reach statistical significance (n = 264 district-years, 11 clusters), and the SPI-based drought and interaction coefficients change sign depending on which canal buffer (2, 5, or 10 km) is used. The hydroclimatic-to-ecological signal here is coherent and well supported; the wheat-yield link is not supported, and that gap is reported as a finding in its own right.

Keywords: drought events; SPI; soil moisture; MODIS; CHIRPS; ERA5-Land; South Punjab; wheat yield; canal irrigation; remote sensing

1. Introduction

South Punjab sits at the arid margin of the Indus Basin irrigation system, and depends on it: wheat, grown in the Rabi season from November to April, is both the dominant crop and a cornerstone of household food security and provincial output. Agricultural outcomes here are really the product of two things interacting — a highly variable hydroclimate and a canal network that is dense in some districts and sparse in others. That makes it practically useful to know how meteorological dryness actually moves through the soil, vegetation, and thermal domains before it reaches the crop, and whether that movement looks different in a canal-dense district than in a marginal one.

This report characterizes drought dynamics in those eleven districts over 2001-2025 using a multi-sensor satellite and reanalysis dataset. It identifies discrete drought events, tests temporal-lag associations along a precipitation-to-soil-to-ecosystem chain, maps a static canal-exposure proxy, and checks district-year drought metrics against an official wheat-yield panel. One caution runs through all of it: association is not causation. The goal is to describe what the data actually show, not to claim a causal drought-propagation mechanism, or an irrigation-buffering effect, that an observational design of this kind cannot establish.

Specific objectives are:

- Construct a consistent multi-sensor monthly district panel (2001-2025) for South Punjab.
- Detect and characterize meteorological (SPI) and soil-moisture drought events using run theory.
- Quantify lagged associations along the drought cascade (precipitation to soil moisture to LST / NDVI / ET).
- Assess Terra-Aqua LST consistency as a data-quality check.
- Relate drought metrics and canal-exposure fraction to wheat yields via panel regression, with robustness checks across buffer distances.

2. Study Area and Period

Eleven districts make up the study area: Multan, Khanewal, Vehari, Lodhran, Bahawalpur, Bahawalnagar, Rahim Yar Khan, Muzaffargarh, Layyah, Dera Ghazi Khan, and Rajanpur — the southern irrigated plains of the Indus Basin. They span a real gradient, both climatic and infrastructural: canal-dense, cropland-intensive districts near the Sutlej and Chenab corridors give way to drier, less canal-exposed districts toward the Sulaiman piedmont (Dera Ghazi Khan, Rajanpur, Layyah). Wheat dominates the Rabi season here; cotton and rice are the main Kharif crops. Section 4.5's yield analysis covers Rabi wheat only.

District boundaries come from FAO GAUL (the 2025 vintage, falling back to 2015 where needed). The study period runs January 2001 through a partial 2025 — whatever months the export covers — with one exception: Aqua MODIS LST only starts in July 2002, since that is when the MYD11A2 record itself begins. Pre-2002 Aqua months are not filled in or estimated; they are simply treated as missing.

Rather than sorting districts into a binary irrigated-versus-rainfed scheme, this study uses a continuous canal-exposure fraction — the share of a district's mapped cropland within a given distance of mapped OSM canals (Section 4.4). That is a more honest fit to what the underlying data can actually support.

3. Data and Methods

Table 1 lists each dataset used, its role in the analysis, and how it was processed.

Table 1. Datasets and processing

Dataset	Native resolution	Role	Period	Processing
CHIRPS V3 (pentad)	~0.05 deg (~5 km)	Precipitation forcing	2001-2025	Aggregated to monthly totals per district; drives SPI-1 / SPI-3
ERA5-Land (monthly aggr.)	0.1 deg (~9-11 km)	Soil-water response	2001-2025	Depth-weighted 0-100 cm volumetric soil water (0.07 / 0.21 / 0.72 layer weights); calendar-month z-scores
MODIS Terra MOD11A2	1 km	Thermal (primary)	2001-2025	Monthly mean daytime LST (scale 0.02, offset -273.15 degC)
MODIS Aqua MYD11A2	1 km	Thermal (consistency check)	Jul 2002-2025	Same processing as Terra; used only to check cross-platform consistency
MODIS MOD13Q1	250 m	Vegetation response	2001-2025	Monthly mean NDVI (scale 0.0001)
MODIS MOD16A2GF	500 m	Evapotranspiration response	2001-2025	8-day composites summed to calendar months (boundary approximation)
ESA WorldCover v200 (2021)	10 m	Cropland mask	Static, 2021	Cropland = class 40; denominator for canal-exposure fraction
OSM canals (vector)	Vector (n/a)	Canal-exposure proxy	Current/static mapping	Cropland within 2 / 5 / 10 km buffer of mapped canal lines
Official wheat panel (CRS/PBS/MNFSR)	District-level (n/a)	Agricultural outcome	Crop years 2000/01-2024/25	District-year yield (t/ha); standardized, detrended residual (YLR)

Note: "Native resolution" is the source product's documented pixel/grid size. It is distinct from the reduction scale used in Earth Engine's reduceRegions() (e.g. 1000 m for LST, 5566 m for CHIRPS); in this pipeline the two happen to coincide for every layer, but that is a property of this specific configuration, not a general guarantee.

3.1 Precipitation and SPI

SPI-1 and SPI-3 come from CHIRPS monthly precipitation, computed with a mixed zero-inflated gamma/normal transform rather than the common shortcut of replacing zero-precipitation months with a small arbitrary constant. The probability of a zero-rainfall month is estimated directly; a gamma distribution is fit to the positive values; the two are combined into one cumulative probability and pushed through the standard normal transform. A negative SPI means a district was drier than its own calendar-

month norm. It is worth being precise about what that means: SPI measures meteorological dryness in precipitation terms, not agricultural drought directly.

3.2 Soil water

Soil water is the ERA5-Land reanalysis depth-weighted volumetric soil water (m3/m3) for the 0-100 cm layer, built from three ERA5-Land depth bands weighted 0.07, 0.21 and 0.72 from shallow to deep. It is a model/reanalysis quantity throughout this report, not a direct satellite or ground observation — that distinction matters and is kept consistent everywhere the variable is used. Anomalies are z-scored within each district and calendar month, so every January month is judged only against other January months in that same district, every July against other July months, and so on; a value of -1 means that month's soil water sat one standard deviation below the district's own historical norm for that month. One side effect is worth flagging: this keeps Rabi months (November-April) from ever being averaged against Kharif months (May-October), which would otherwise blur two seasons with very different baseline soil-water levels. What it does not do is remove any slow trend within a given calendar month across the 25-year record — say, from gradually changing irrigation practices. No separate detrending step was applied to these monthly anomalies (see Limitations).

3.3 Thermal, vegetation, and ET

Terra's MOD11A2 daytime LST is the primary thermal indicator. Aqua's MYD11A2 LST, available from July 2002, plays a narrower role: a cross-platform consistency check (Section 4.1), not independent validation, since Terra and Aqua are both MODIS instruments rather than separate systems. NDVI is from MOD13Q1, ET from MOD16A2GF, both aggregated from 8-day composites up to calendar months — an approximation at the month boundaries worth keeping in mind. VCI and TCI were also computed, but only as descriptive, supporting indicators; neither is treated as a standalone drought index.

3.4 Drought event detection

Events are detected on the standardized anomaly series — SPI-1 and the soil-water z-score — using run theory with a single-month gap-pooling rule: a run below threshold can absorb one intervening non-drought month, provided the month after that is below threshold again. Minimum duration is two months. Three thresholds were tested: -0.5 (mild), -1.0 (the primary, moderate category), and -1.5 (severe). Unless stated otherwise, every event count and comparison in this report uses the -1.0 threshold. Duration is inclusive (end minus onset, plus one month), and peak intensity is simply the most negative value reached during the event.

Cumulative severity is S = sum of max(0, tau - Z_t) over the event's months, where tau is the threshold, and Z_t is the standardized anomaly in month t. The max(0, .) matters: it means a bridged gap month, where Z_t sits above the threshold, adds nothing to the sum — only months actually below the threshold count toward severity. That matches exactly what the pipeline computes. A simpler S = sum(tau - Z_t) would get this wrong, letting an above-threshold gap month quietly subtract from the running total.

3.5 Lag associations

For each district, Spearman correlations were run between deseasonalized anomaly pairs at lags of 0-3 months, across four relationships: precipitation drought to soil water, and soil water to thermal, NDVI, and ET anomalies. The precipitation metric here is SPI-1, deliberately not SPI-3 — a three-month rolling accumulation would smear any lagged response across the lag window itself and muddy the estimate. SPI-3 still shows up later, but only as a separate robustness check on the wheat-yield regression (Model C, Section 4.5), where lag timing is not the question. Benjamini-Hochberg FDR correction was applied across the full set of tests. None of this should be read as causal propagation testing — these are exploratory associations on autocorrelated monthly series, so direction, magnitude, and how consistent a correlation is across districts (Section 4.3) matter more here than the p-value on its own.

3.6 Canal exposure

Canal exposure is the fraction of a district's ESA WorldCover cropland (class 40) within a given buffer of a mapped OSM canal line, with total district cropland area as the denominator:

> *canal_exposure_fraction = (cropland area within B km of mapped canals) / (total cropland area in the district)*

B = 5 km is primary; 2 km and 10 km were also run for sensitivity. This is a static snapshot — 2021 land cover, current canal mapping — and it measures proximity, nothing more. It is not irrigation percentage, not irrigated area, not water delivery. A field can sit inside the buffer and still go dry in a given season; all this proxy says is how close, on average, a district's cropland sits to mapped canal infrastructure.

3.7 Wheat yield and the yield regression

The wheat panel (crop years 2000/01-2024/25, t/ha) is an official composite, not the standalone Kaggle dataset. It stitches together the Crop Reporting Service panel for 2001/02-2008/09, Pakistan Bureau of Statistics figures for 2009/10-2019/20, and Ministry of National Food Security and Research data (with CRS) for 2021/22-2024/25. Two single years per district do not have a clean official figure — 2000/01 and 2020/21 — and were filled using a Kaggle-CRS blend; out of 25 years per district, those two are the only ones where Kaggle contributes anything. Yield is then regressed on a linear time trend for each district, and the standardized residual from that fit is the Yield-Level Residual (YLR) used throughout. A negative YLR simply means that year's yield came in below the district's own long-run trend — it is not a drought-attribution claim, and not a percentage-loss figure. Rabi-season (November-April) drought metrics are matched to the harvest year the season feeds into, so November-December 2022 and January-April 2023 both count toward the 2023 harvest. A district-year only makes it into the panel if at least 5 of its 6 Rabi months have valid soil-moisture data.

The regression model is:

> *YLR_it = beta1 * drought_it + beta2 * (drought_it x canal_exposure_i) + district fixed effects + harvest-year fixed effects + error_it*

Three drought metrics were pre-specified as separate robustness checks, not as a specification search to pick the best-looking one: (A) mean Rabi soil-moisture z-score, (B) mean Rabi SPI-1, (C) mean Rabi SPI-3. Standard errors are clustered by district. With only 11 clusters to work with, that clustering is treated as indicative throughout Sections 4.5 and 5 — not as formal confirmatory inference.

4. Results

4.1 Data coverage and Terra-Aqua consistency check

The merged monthly panel runs the full 2001-2025 window for precipitation, soil water, Terra LST, NDVI and ET across all 11 districts; Aqua LST joins from July 2002. Over the 3,102 district-months where Terra and Aqua overlap, the two track each other closely (Pearson r = 0.9909) but are not identical — Terra's daytime LST sits a mean of 3.282 degC (SD 1.045 degC) below Aqua's, which makes sense given Terra's earlier overpass (~10:30 local, versus Aqua's ~13:30) catches a cooler part of the day.

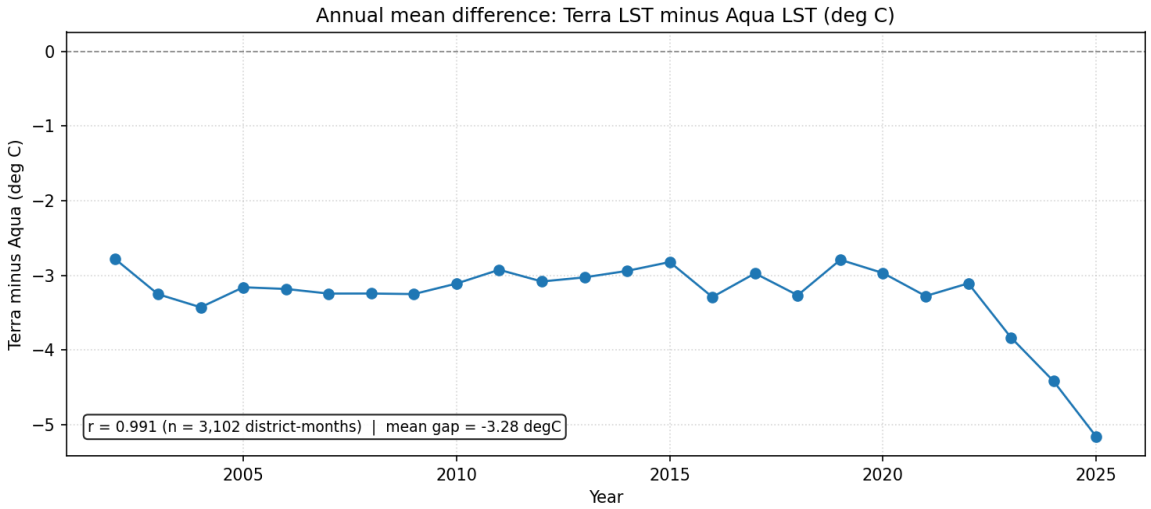


Figure 1. Annual mean Terra-minus-Aqua daytime LST difference, 2002-2025 (n = 3,102 overlapping district-months; r = 0.9909).

That gap is not constant, either. A linear fit to the annual means gives a slope of -0.0343 degC per year, but the real story is in the last three years: -3.831 degC in 2023, -4.417 in 2024, -5.159 in 2025 — well outside the roughly -2.8 to -3.4 degC band that held from 2002 through 2022. Worth being clear about what this is and is not: it is a sensor-consistency check, not validation of one product against the other (Terra and Aqua are related MODIS instruments, not independent ground truth), and the pattern in the data is a sharp recent widening rather than a steady drift since 2020. Anyone drawing Terra/Aqua comparisons from 2023-2025 specifically should know that.

4.2 Drought event characteristics

At the -1.0 threshold, the 11 districts produced 91 SPI-1 (precipitation) events and 48 soil-moisture events over 2001-2025. The two behave differently: SPI-1 events are more numerous, shorter, and spread fairly evenly across districts and years, while soil-moisture events are rarer but run longer — in several districts, stretching across multiple consecutive years.

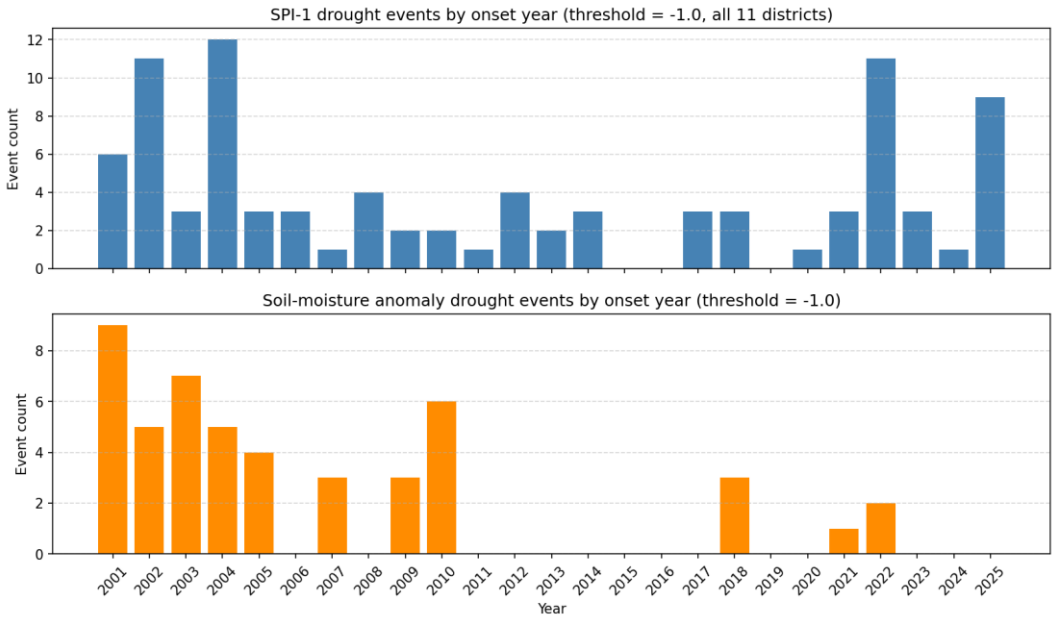


Figure 2. SPI-1 and soil-moisture drought events by onset year, threshold = -1.0, all 11 districts.

SPI-1 onsets cluster in 2002, 2004, 2022, and early 2025 — 11, 12, 11, and 9 of the 91 total, respectively — which lines up with dry years widely reported across the region. Soil-moisture onsets cluster differently, concentrated in 2001 (9) and 2003 (7): soil water reflects accumulated deficit, not any single dry month, so its peaks do not always land on the same years as SPI-1's.

Table 2 ranks the ten most severe events by cumulative severity, pooling both indicators. Nine of the ten are soil-moisture events, not SPI-1 — and seven of the ten sit in just three districts: Dera Ghazi Khan (3), Bahawalpur (2), and Rajanpur (2), with Bahawalnagar, Rahim Yar Khan, and Layyah accounting for the rest. These events also cluster in time, mostly in 2001-2004 and again in 2007-2008.

Table 2. Ten most severe drought events by cumulative severity, 2001-2025

District	Indicator	Onset	End	Duration (mo)	Cum. severity	Peak intensity
Bahawalpur	SM_z	2002-01	2003-06	18	9.358	-1.854
Dera Ghazi Khan	SM_z	2001-11	2003-06	20	9.091	-1.903
Dera Ghazi Khan	SM_z	2003-10	2004-12	15	6.806	-1.920
Rajanpur	SM_z	2001-11	2003-06	20	6.600	-1.664
Bahawalpur	SM_z	2007-04	2008-07	16	5.473	-1.806
Rajanpur	SM_z	2003-10	2004-12	15	5.199	-1.735
Bahawalnagar	SM_z	2002-01	2003-01	13	3.473	-1.539
Rahim Yar Khan	SM_z	2005-11	2006-07	9	3.301	-1.686
Dera Ghazi Khan	SPI1	2020-09	2021-02	6	3.229	-3.464
Layyah	SM_z	2003-11	2004-12	14	3.185	-1.654

That pattern — concentrated in Dera Ghazi Khan, Rajanpur, and Bahawalpur, particularly during 2001–2004 and 2007–2008 — comes back in the Discussion. It suggests the record's most severe, sustained soil-water deficits are not spread evenly across South Punjab or across the 25 years; they cluster in Dera Ghazi Khan, Rajanpur and Bahawalpur, and in two specific windows.

4.3 Multi-sensor lag associations

Figure 3 presents Multan's monthly record, 2001-2025, with SPI-1, the soil-water anomaly, and NDVI, thermal and ET anomalies on the same time axis. Soil-water deficits are associated with lower NDVI and ET and higher surface temperature. The district-mean statistics in Table 3 quantify those different signs; the time series provides context rather than causal attribution.

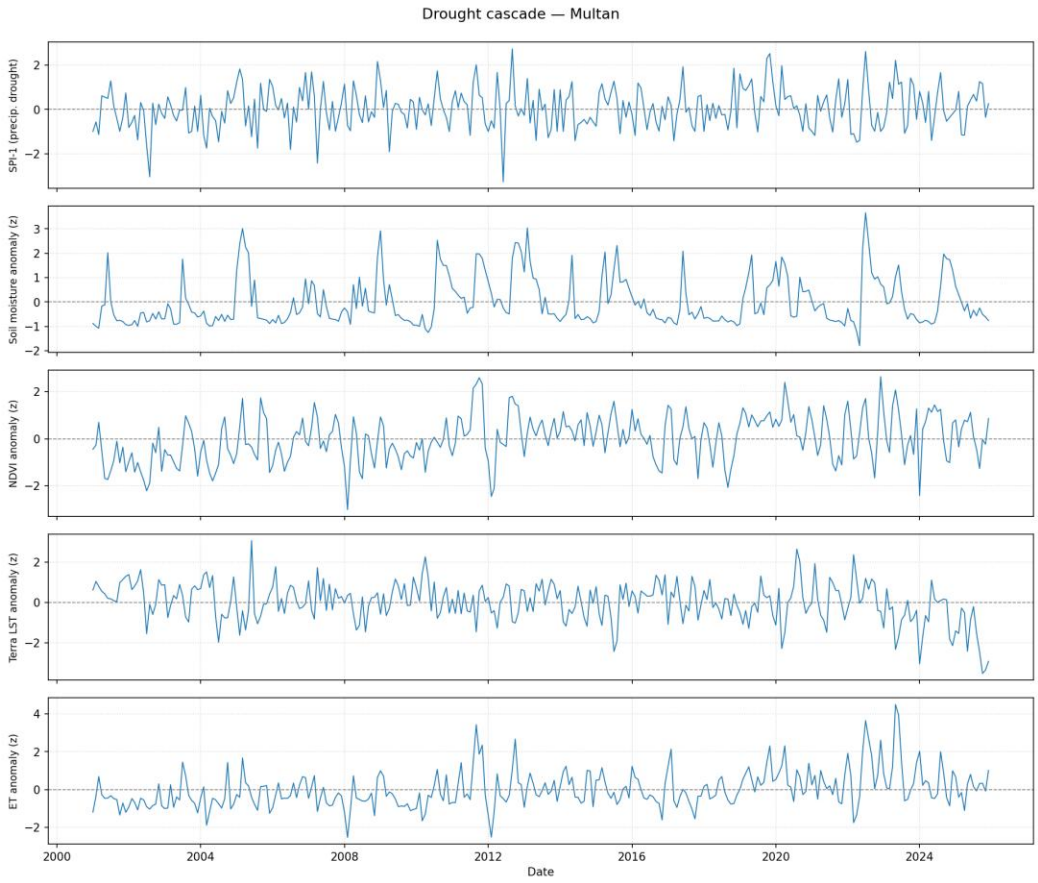


Figure 3. Drought cascade for Multan: SPI-1, soil-moisture anomaly, NDVI anomaly, Terra LST anomaly, and ET anomaly, 2001-2025 (all as z-scores except SPI-1).

Table 3 and Figure 4 give the district-mean Spearman correlations for all four relationships at lags 0-3, across all 11 districts. Every one of them peaks at lag 0 and decays as the lag grows — none shows a delayed peak. And at lag 0, every relationship is FDR-significant in all 11 districts, all agreeing in sign with the district mean.

Table 3. District-mean lag associations (Spearman r, deseasonalized anomalies)

Relationship	Lag 0	Lag 1	Lag 2	Lag 3	Districts sig. at peak
Precipitation (SPI-1) -> Soil water	0.288	0.247	0.148	0.089	11 / 11
Soil water -> Thermal (LST)	-0.292	-0.167	-0.116	-0.064	11 / 11
Soil water -> NDVI	0.483	0.398	0.320	0.300	11 / 11
Soil water -> ET	0.465	0.293	0.179	0.134	11 / 11

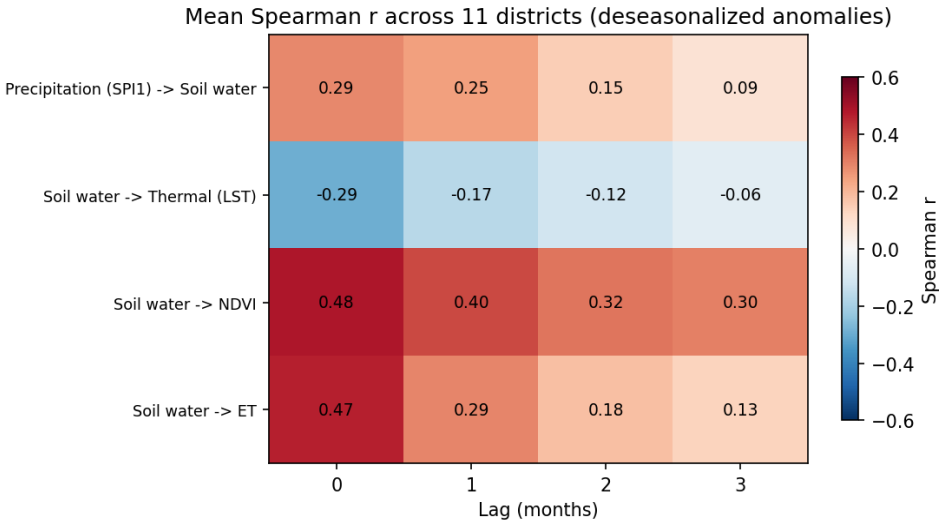


Figure 4. Mean Spearman r across 11 districts, by relationship and lag (months).

Soil water to NDVI and soil water to ET are the strongest pair of the four — peak r of 0.483 and 0.465 — and the most durable, both still above 0.30 and 0.13 at three months out. Soil water to thermal runs negative, as expected (drier soil, hotter surface), but it is weaker and fades faster. Precipitation to soil water sits in the middle at lag 0 and drops off quickly, which tracks: at monthly district resolution, soil water responds to rain within the same month rather than months later.

That everything peaks at lag 0, rather than later, is itself the more conservative and more informative result. It says the strongest signal here is contemporaneous covariation with a persistence tail — not a multi-month cascade where the response only shows up after the driver has already passed. Vegetation, thermal and ET anomalies co-move with soil water and fade slowly over the following one to three months; that is different from saying soil-water deficit causes ecosystem stress at some fixed lag, and the data do not support the stronger claim.

4.4 Canal-exposure geography

At the primary 5 km buffer, canal exposure swings from 0.452 in Layyah to 0.927 in Bahawalnagar — almost half a unit of range across just eleven districts. Table 4 breaks this out at all three buffers tested. The ranking of districts holds up fairly well across 2, 5, and 10 km, but the spread compresses as the buffer widens: by 10 km, Khanewal, Lodhran, Vehari, Multan, Bahawalpur and Bahawalnagar have all crossed 0.9, while Layyah and Dera Ghazi Khan are still comparatively low, at 0.69 and 0.85.

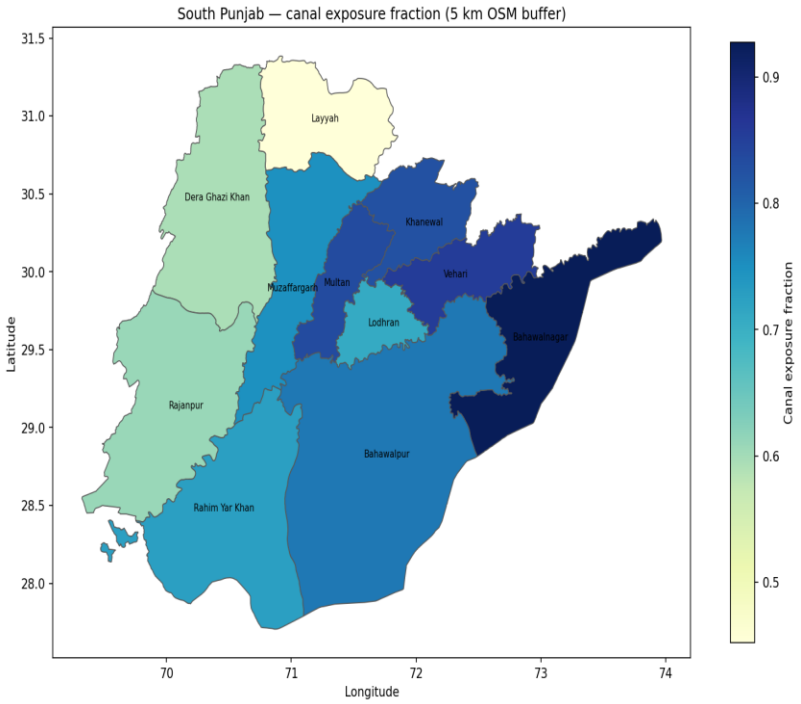


Figure 5. Canal-exposure fraction (5 km OSM buffer, cropland denominator) by district.

Table 4. Canal-exposure fraction by district, across buffer distances

District	2 km	5 km (primary)	10 km
Bahawalnagar	62.79%	92.71%	98.98%
Vehari	49.48%	85.58%	99.94%
Multan	48.47%	83.55%	99.00%
Khanewal	41.57%	82.65%	100.00%
Bahawalpur	39.52%	77.84%	97.67%
Muzaffargarh	43.98%	74.83%	92.98%
Rahim Yar Khan	45.70%	72.66%	91.52%
Lodhran	35.92%	70.94%	99.71%
Rajanpur	35.80%	60.98%	83.98%
Dera Ghazi Khan	28.62%	59.36%	84.81%
Layyah	24.60%	45.23%	69.02%

Rows are sorted by the 5 km value. Layyah, Dera Ghazi Khan, and Rajanpur have the lowest canal exposure at every buffer tested and account for six of the ten severe events in Table 2. This is a descriptive spatial overlap, not evidence that canal access caused differences in drought severity. The yield regressions in Section 4.5 address agricultural associations rather than directly testing this event-geography pattern.

4.5 Wheat-yield comparison and robustness

The eligible Rabi panel has 264 district-year observations across 11 districts and harvest years 2002-2025. The documented minimum of five valid soil-moisture months is now enforced before merging and fitting the models. Harvest year 2001 has only January-April within the environmental record and is excluded. All retained seasons have six soil-moisture months. Table 5 reports the three prespecified models with district and harvest-year fixed effects.

Table 5. Wheat-yield regression: drought and canal-exposure interaction (primary, 5 km buffer)

Model	Drought metric	Drought coef. (SE)	p	Interaction coef. (SE)	p	Adj. R^2
A	Rabi mean SM z-score	-0.456 (0.409)	0.265	0.481 (0.547)	0.380	0.384
B	Rabi mean SPI-1	0.354 (0.806)	0.661	0.168 (1.238)	0.892	0.384
C	Rabi mean SPI-3	0.031 (0.542)	0.955	0.182 (0.945)	0.848	0.379

n = 264 district-years, 11 clusters, in all three specifications; standard errors are clustered by district, and with only 11 clusters the resulting p-values are indicative rather than confirmatory.

None of the three drought coefficients or interaction terms reaches the conventional 0.05 significance threshold. The drought p-values are 0.265, 0.661 and 0.955; interaction p-values are 0.380, 0.892 and 0.848. Main drought coefficients are negative for soil moisture and positive for both SPI metrics, while all three interactions are positive at the primary buffer. Adjusted R-squared ranges from 0.379 to 0.384. These exploratory estimates do not establish a drought-yield or irrigation-buffering effect.

Buffer sensitivity (2 / 5 / 10 km)

The 2, 5 and 10 km sensitivity results are reported in Appendix Table 6. Model A's interaction remains positive (0.685, 0.481 and 0.390 respectively); Models B and C have positive interactions at 2 and 5 km and negative interactions at 10 km. The SPI-based drought coefficients also change sign across buffers. None of the drought or interaction terms reaches the conventional 0.05 threshold in these nine fits. With only 11 clusters and coefficients that depend on the exposure definition, these agricultural results remain exploratory and inconclusive.

5. Discussion

Section 4's multi-sensor evidence adds up to a coherent picture of hydroclimatic drought in South Punjab. Precipitation deficits, soil-water deficits, and land-surface thermal, vegetation and ET anomalies all co-vary in the expected directions, strongest contemporaneously and fading gradually over one to three months. That is a genuinely strong result at the descriptive and associative level. It does not amount to a causal propagation chain: the lag-0 peak in every single relationship means this analysis does not provide evidence for a distinctly lagged, multi-month cascade at the monthly district scale. That is a narrower

claim than saying no such cascade exists — monthly aggregation, serial autocorrelation, and drivers that move together in time could all make a genuinely delayed physical response look contemporaneous at this resolution — but it is the claim the data actually support, and no stronger one is made here.

This study really asks two separate questions, and they get different answers. Does precipitation deficit track soil-water deficit, and does soil-water deficit track thermal, vegetation and ET stress? Yes — 25 years of dense, multi-sensor, district-level data back that up, with consistent signs and FDR significance in all 11 districts. Does that drought signal actually show up in wheat-yield residuals, and does canal proximity change the picture? No, not in this data: none of the three drought metrics, at any of the three buffers, produces a stable, significant association with the yield residual.

That gap deserves real discussion, not a shrug. Wheat yield depends on far more than a district-average Rabi drought metric — cultivar, fertilizer, pest pressure, mechanization, market conditions, field-level water access that varies within a district and never shows up at this resolution. Any of those could be diluting a real drought signal down below detectability, especially with only 11 clusters to work with. It is also entirely possible the true effect is just small, or depends heavily on conditions this study cannot see. Nothing here lets us pick between those explanations, and this report is not going to pretend otherwise.

Layyah, Dera Ghazi Khan and Rajanpur combine low canal exposure with six of the ten severe events, but the spatial overlap does not identify a buffering mechanism. These districts differ in climate, cropland and farming conditions. The district-level yield models do not establish an agricultural buffering effect, and they do not directly test whether canal access explains the geography of severe events.

There is a deeper issue here too, about how a buffering hypothesis should even be tested. A 'green NDVI next to dry soil' pattern is not, by itself, proof of canal irrigation decoupling vegetation from rainfall — canal seepage and rainfall wet the same soil, and neither ERA5-Land nor NDVI can tell the two apart in a given pixel. A sharper test would compare how NDVI and ET respond to precipitation anomalies specifically, not to the soil-water anomaly (which canal water already contaminates), between high- and low-exposure districts: do canal-dense districts show a muted response to a given rainfall deficit compared to the rainfed margin? This study's district-year interaction term is a blunter version of that test, and it came up empty (Section 4.5) — but a null result at this resolution could mean several things besides no buffering at all: differing crop calendars between districts, cloud contamination in the optical NDVI signal during monsoon months, or simple pixel mixing, since a single ~9-11 km ERA5-Land soil-water cell can blend canal-adjacent cropland with rainfed land next door. One more thing worth saying plainly: canal supply itself is not independent of the basin-wide conditions this study calls 'drought.' Canal discharge in the Indus system is drought-sensitive too, so sitting close to a canal is no guarantee of a stable water supply during a regional dry spell — and this study has no discharge or off-take data to check that separately.

One more caution on data quality: the Terra-Aqua gap widens noticeably in the last three years (Section 4.1). The two sensors are still very highly correlated overall, but anyone drawing on 2023-2025 specifically for Terra/Aqua comparisons should know the offset in that window — roughly -3.8 to -5.2 degC — runs well past the ~25-year average of -3.28 degC.

Three things could push the agricultural-linkage question further than this study's inputs allow. Higher-resolution soil moisture — SMAP, or Sentinel-1 backscatter — would cut down the ~9-11 km pixel mixing flagged above. A dynamic, within-season irrigation signal, whether a microwave irrigated-area map or an NDVI-based cropping-intensity measure, would let canal exposure move year to year instead of sitting fixed at its 2021 snapshot. And crop-model assimilation, or field-level yield records where they exist, would let a drought-yield test actually control for management factors — cultivar, fertilizer, sowing date — that a district-year aggregate like this one simply cannot separate from drought.

6. Limitations

- All variables are aggregated to the district level, which masks within-district heterogeneity (a single district can contain both heavily canal-exposed and nearly rainfed cropland, averaged into one monthly value).
- ERA5-Land soil water is a model/reanalysis product, not a direct in-situ or satellite retrieval of soil moisture.
- Canal exposure is a static (2021 WorldCover / current OSM) infrastructure-proximity proxy: it does not track land-use or cropland change over the 25-year study period, is not a measure of actual annual irrigation-water delivery, and canal supply itself is not independent of the same basin-wide drought conditions being studied (see Section 5).
- Wheat-yield inference rests on only 11 independent district clusters; cluster-robust standard errors at this scale are indicative, not confirmatory.
- The Yield-Level Residual (YLR) reflects deviation from a district-specific linear time trend, not a proven drought-caused yield loss; alternative detrending choices could give different residuals.
- Calendar-month z-scoring keeps Rabi and Kharif months from being pooled together, but does not detrend for long-run changes within a given month across the 25-year record (e.g., changing irrigation practice or cropping intensity).
- Aqua MODIS LST is only available from July 2002 onward, about 18 months later than the rest of the record.
- ET (MOD16A2GF) is one of the four core response layers used throughout this study, not a planned future addition; it is aggregated from 8-day composites to calendar months, which introduces a month-boundary approximation. An alternative ET product (e.g. ETMonitor or PML_V2) could in principle be substituted or cross-checked, but this study did not verify the access terms, coverage, or licensing of any alternative ET source, so no such comparison is reported here.
- OSM canal mapping completeness and positional accuracy are not independently verified and may vary across the study area.
- Lag correlations are exploratory, not causal-propagation tests; the underlying monthly series are autocorrelated, which affects how their p-values should be read.

- District and harvest-year fixed effects absorb a large share of variation in the yield regression, leaving limited residual variation for the drought predictors to explain.
- Pakistan's August-September 2022 monsoon floods caused severe, well-documented damage specifically in Dera Ghazi Khan and Rajanpur districts (hill-torrent flooding, hundreds of villages and tens of thousands of houses affected) — two of the three lowest-canal-exposure districts highlighted in this report. This study does not include a flood-exposure control, so the 2023 harvest-year drought and yield metrics for these two districts in particular should be interpreted with additional caution, since post-flood soil and cropping conditions are not separable from the drought metrics used here.
- The observational design of the whole study does not establish causal effects of drought, or of canal irrigation, on yield.

7. Conclusion

The study supports the following conclusions, each stated at the level of evidence the data actually provide:

- A 25-year, district-month multi-sensor panel for South Punjab (precipitation, soil water, LST, NDVI, ET, canal exposure) was constructed and quality-checked, including an internal Terra/Aqua consistency check ($r = 0.991$, $n = 3{,}102$ overlapping district-months).
- 91 SPI-1 and 48 soil-moisture drought events (threshold -1.0) were identified and characterized by duration, severity, and timing across the 11 districts; the most severe, longest events cluster in Dera Ghazi Khan, Rajanpur, and Bahawalpur, particularly during 2001-2004 and 2007-2008.
- Precipitation, soil-water, thermal, vegetation, and ET anomalies show statistically consistent, same-sign associations across all 11 districts, strongest at a contemporaneous (zero-month) lag and persisting in attenuated form for one to three months — a coherent hydroclimatic-to-ecological signal, not evidence of a delayed multi-month propagation sequence.
- Canal-exposure fraction is spatially heterogeneous (0.452 to 0.927 across the 11 districts) and is descriptively lowest in the same districts that experienced the most severe recorded soil-moisture droughts, but this study evaluated that pattern as a spatial hypothesis, not as proof of an irrigation-buffering mechanism.
- District- and harvest-year fixed-effects regression of standardized detrended wheat-yield residuals on Rabi-season drought metrics and their interaction with canal exposure showed no statistically clear association in any of three pre-specified drought metrics, and the sign and size of both the drought and interaction coefficients were unstable across 2, 5, and 10 km buffer definitions.
- The clearest, best-supported contribution of this study is the multi-sensor hydroclimatic characterization; the wheat-yield linkage is reported as a genuinely inconclusive exploratory result rather than as either a confirmed drought-yield relationship or a confirmed null effect.

Acknowledgements

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Appendix

Table 6. Wheat-yield regression buffer sensitivity (2 / 5 / 10 km), all three drought metrics

Buffer	Model	Drought coef. (SE)	p	Interaction coef. (SE)	p	Adj. R^2
2 km	A (SM z)	-0.390 (0.243)	0.108	0.685 (0.522)	0.189	0.385
2 km	B (SPI-1)	-0.043 (0.600)	0.943	1.256 (1.505)	0.404	0.388
2 km	C (SPI-3)	-0.184 (0.365)	0.614	0.896 (1.165)	0.442	0.383
5 km	A (SM z)	-0.456 (0.409)	0.265	0.481 (0.547)	0.380	0.384
5 km	B (SPI-1)	0.354 (0.806)	0.661	0.168 (1.238)	0.892	0.384
5 km	C (SPI-3)	0.031 (0.542)	0.955	0.182 (0.945)	0.848	0.379
10 km	A (SM z)	-0.464 (0.773)	0.548	0.390 (0.837)	0.641	0.382
10 km	B (SPI-1)	1.283 (0.950)	0.177	-0.909 (1.173)	0.439	0.385
10 km	C (SPI-3)	0.605 (0.717)	0.399	-0.520 (0.927)	0.575	0.380

n = 264 district-years (11 district clusters) in all nine rows; standard errors (in parentheses) are clustered by district. No coefficient in any of the nine rows reaches conventional significance.

Table 7. Per-district lag-0 Spearman r underlying the "significant in all 11 districts" claim (Section 4.3)

Section 4.3 and the Abstract both claim all four lag relationships are FDR-significant in all 11 districts at lag 0. Table 3 only shows the district mean, so the full district-level breakdown is reproduced below, straight from lag_correlations_anomaly_based.csv — the claim should be checkable here without reopening that file.

District	Precip. -> Soil water	Soil water -> Thermal	Soil water -> NDVI	Soil water -> ET
Bahawalnagar	0.294	-0.319	0.556	0.498
Bahawalpur	0.125	-0.180	0.697	0.396
Dera Ghazi Khan	0.260	-0.356	0.748	0.597
Khanewal	0.370	-0.256	0.305	0.452
Layyah	0.293	-0.268	0.482	0.481
Lodhran	0.360	-0.172	0.203	0.372
Multan	0.355	-0.280	0.337	0.451
Muzaffargarh	0.332	-0.416	0.515	0.535
Rahim Yar Khan	0.212	-0.386	0.574	0.458
Rajanpur	0.164	-0.361	0.641	0.437
Vehari	0.408	-0.217	0.255	0.442

All 44 values here (11 districts times 4 relationships) are individually FDR-significant at lag 0. Even the weakest one — Lodhran's soil-water-to-NDVI correlation, r = 0.203 — carries p_fdr = 7.39e-4.

Data and output locations

The file names below identify the source tables. Current corrected wheat tables are in the November release and public repository; the direct Drive links retain the original archive. Environmental results are unchanged. See the Revised analysis release note for the 264-row correction.

- Table 1 (dataset resolutions and roles) - documented native product specifications; not derived from a single export file.
- Table 2 (top-10 severe events) and Figure 2 (event timelines) - SPI1_drought_events.csv and SM_drought_events.csv (analysis_outputs/ has the primary-threshold subsets: SPI1_events_threshold_m1p0.csv, SM_events_threshold_m1p0.csv), and major_drought_events_top10.csv.
- Figure 3 (Multan cascade) - figures/cascade_timeseries_Multan.png, built from south_punjab_drought_analysis_panel.csv.
- Table 3, Figure 4, and Table 7 (lag associations) - lag_correlations_anomaly_based.csv (district x pair x lag; Table 3 is the district-mean of that file, Table 7 is the individual district-level values at lag 0).
- Table 4 and Figure 5 (canal exposure) - gee_exports/district_canal_exposure_fraction.csv (5 km primary), _2km.csv, and _10km.csv.
- Table 5 (primary yield regression) - yield_regression_robustness_summary.csv; underlying panel in wheat_drought_validation_panel.csv (264 rows).
- Table 6 / Appendix (buffer sensitivity) - yield_regression_buffer_sensitivity.csv.
- Section 4.1 (Terra/Aqua) - terra_aqua_consistency.csv and terra_aqua_annual_gap.csv.
- south_punjab_drought_analysis_panel.csv - the full monthly district panel with all derived anomalies, SPI, and event flags, underlying nearly every table and figure above.

Direct links

Every file and folder named above, with a direct Google Drive link:

File / folder	Google Drive link
AiRCAS_drought/ (project root)	Open
analysis_outputs/	Open
gee_exports/	Open
figures/	Open
south_punjab_drought_analysis_panel.csv	Open
SPI1_drought_events.csv	Open
SM_drought_events.csv	Open
major_drought_events_top10.csv	Open

File / folder	Google Drive link
lag_correlations_anomaly_based.csv	Open
yield_regression_robustness_summary.csv	Open
wheat_drought_validation_panel.csv	Open
yield_regression_buffer_sensitivity.csv	Open
terra_aqua_consistency.csv	Open
terra_aqua_annual_gap.csv	Open
figures/cascade_timeseries_Multan.png	Open
figures/canal_exposure_map.png	Open
gee_exports/SPI1_events_threshold_m1p0.csv	Open
gee_exports/SM_events_threshold_m1p0.csv	Open
gee_exports/district_canal_exposure_fraction.csv (5 km)	Open
gee_exports/district_canal_exposure_fraction_2km.csv	Open
gee_exports/district_canal_exposure_fraction_10km.csv	Open
source_data/south_punjab_wheat_official_2000_2024_with_source.csv (Sec. 3.7 provenance)	Open

Reproducibility

The full pipeline runs as a single Colab notebook in two passes, controlled by two flags: an export pass (RUN_GEE_EXPORTS = True) that authenticates to Google Earth Engine and starts the Drive CSV export tasks listed in Table 1, and an analysis pass (RUN_ANALYSIS = True) that collects the completed exports, computes SPI, soil-moisture z-scores, event catalogs, lag correlations, Terra-Aqua diagnostics, and the yield regressions, then writes the CSV outputs and PNG figures referenced throughout this report. The notebook, source CSVs, OSM canal GeoJSON, and all derived products are archived in the project's AiRCAS_drought Google Drive folder; the analysis can be re-run end-to-end once the export tasks have completed.

Revised analysis release

The original analysis was rerun from the preserved Drive exports; all 12 original CSV outputs matched within numerical tolerance. This edition then enforces the report's five-month Rabi completeness rule, excluding 11 incomplete first-season records. Tables 5 and 6 and the sample-size statements use the corrected 264-row panel. Event catalogs and environmental statistics are unchanged. The original September report and archived outputs remain preserved.
Current corrected tables, code, source hashes and reproduction instructions: https://github.com/musab-shahid/south-punjab-drought-analysis . Original Drive links above refer to the historical archive. The source-tagged wheat table contains 22 gap-filled records (2000/01 and 2020/21); 11 remain in the eligible regression panel. Source labels document the compilation but are not an independent certification of each underlying statistical publication.